

1. $1 \text{ AU} = 1.5 \times 10^{11} \text{ m}$
2. $1 \text{ ly} = 9.46 \times 10^{15} \text{ m} = 6.31 \times 10^4 \text{ AU}$
3. $R_{\odot} = 6.96 \times 10^8 \text{ m}$
4. $R_{\oplus} = 6,370 \text{ km} = 6.37 \times 10^6 \text{ m}$
5. $d_{\text{pluto}} = 40 \text{ AU}$
6. $d_{\text{proxima}} = 4.3 \text{ ly}$
7. $T = 10^{12}, G = 10^9, M = 10^6, k = 10^3, m = 10^{-3}, \mu = 10^{-6}, n = 10^{-9}, p = 10^{-12}, a = 10^{-15}$
8. $M_{\odot} = 2.0 \times 10^{30} \text{ kg}$
9. $M_{\oplus} = 5.97 \times 10^{24} \text{ kg}$
10. Circle: $A = \pi r^2, C = 2\pi r$
11. Sphere: $V = \frac{4}{3}\pi r^3, \text{SA} = 4\pi r^2$
12. Rectangle: $A = lw, P = 2l + 2w$
13. Rectangular Prism: $V = lwh, \text{SA} = 2lw + 2wh + 2hl$
14. $v = d/t$
15. $c = 3 \times 10^8 \text{ m/s}$
16. $1 \text{ light-year (1 ly)} = 9.46 \times 10^{15} \text{ m}$
17. $t_{\text{universe}} = 13.8 \text{ Gyr}$
18. $t_{\text{solarsystem}} = 4.56 \text{ Gyr}$
19. $t_{\text{orbit Milky Way}} = 240$
20. $v_{\text{orb}} = \frac{2\pi r}{t}$

$$\mathbf{21.} \quad \sin(\theta) = \frac{\text{opp}}{\text{hyp}}$$

$$\mathbf{22.} \quad \cos(\theta) = \frac{\text{adj}}{\text{hyp}}$$

$$\mathbf{23.} \quad \tan(\theta) = \frac{\text{opp}}{\text{adj}}$$

$$\mathbf{24.} \quad G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$$

$$\mathbf{25.} \quad g = 9.8 \text{ m/s}^2 \text{ (Earth)}$$

$$\mathbf{26.} \quad F = ma$$

$$\mathbf{27.} \quad F_g = mg$$

$$\mathbf{28.} \quad F_g = \frac{GMm}{r^2}$$

$$\mathbf{29.} \quad g = \frac{GM}{r^2}$$

$$\mathbf{30.} \quad I = mr^2$$

$$\mathbf{31.} \quad I\omega_1 = I\omega_2$$

$$\mathbf{32.} \quad \omega = \frac{360^\circ}{t} \text{ (spin rate)}$$